

- Due to *identical particles* assume in quantum mechanics, it does not make sense to *number* the particles.
Therefore, It is convenient to calculate in many-body system to replace the coordinate representation with the *occupation particle number representation*.
- For occupation particle number representation, the type of particles (Boson or Fermion) can be completely determined by the commutation relation of *creation* and *annihilation* operators.

Particle number representation and creation annihilation operator

1. We start from the one-dimension *harmonic oscillator model*. Its Hamiltonian reads $H = \frac{1}{2}p^2 + \frac{1}{2}x^2$

Here, we directly introduce the *creation and annihilation* as $a^\dagger = \frac{1}{\sqrt{2}}(x - ip)$, $a = \frac{1}{\sqrt{2}}(x + ip)$

We can get its commutation relation with condition $[x, p] = xp - px = i$ as $[a, a^\dagger] = 1$

$$aa^\dagger - a^\dagger a = \frac{1}{2}(\textcolor{red}{x}^2 - \textcolor{blue}{i}xp + \textcolor{blue}{i}px + \textcolor{green}{p}^2 - \textcolor{red}{x}^2 - \textcolor{blue}{i}xp + \textcolor{blue}{i}px - \textcolor{green}{p}^2) = -i(xp - px) = 1$$

2. Then by $(a + a^\dagger)$ and $(a - a^\dagger)$, we can get the x, p with its representation $x = \frac{1}{\sqrt{2}}(a^\dagger + a)$, $p = \frac{1}{\sqrt{2}}i(a^\dagger - a)$

Then the Hamiltonian can be read $\hat{H} = \frac{1}{2}p^2 + \frac{1}{2}x^2 = \hat{N} + \frac{1}{2}$ with condition $[a, a^\dagger] = 1$

$$\begin{aligned} & -\frac{1}{4}(a^\dagger - a)(a^\dagger - a) + \frac{1}{4}(a^\dagger + a)(a^\dagger + a) \\ &= \frac{1}{4}(-\textcolor{red}{a}^\dagger \textcolor{red}{a}^\dagger + a^\dagger a + a a^\dagger - \textcolor{green}{a} a + \textcolor{red}{a}^\dagger \textcolor{red}{a}^\dagger + a^\dagger a + a a^\dagger + \textcolor{green}{a} a) \\ &= \frac{1}{2}(a^\dagger a + a a^\dagger) = \frac{1}{2}(2a a^\dagger + 1) = \hat{N} + \frac{1}{2} \end{aligned}$$

3. The $\hat{N}$ is Hermitian operator namely *particle number operator*, its eigenequation can be written as $\hat{N}|n\rangle = n|n\rangle$

The particle number operator can be rewritten as $\hat{N} = a^\dagger a$ with $a|n\rangle = \sqrt{n}|n-1\rangle$, $a^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle$

Therefore, the eigenequation can be described by $\hat{N}|n\rangle = a^\dagger a|n\rangle = a^\dagger \sqrt{n}|n-1\rangle = \sqrt{n}\sqrt{n}|n\rangle = n|n\rangle$

The *particle number eigenstate* $|n\rangle = \frac{1}{\sqrt{n!}} (a^\dagger)^n |0\rangle$

$$|n\rangle = a^\dagger a^\dagger \dots a^\dagger |0\rangle = a^\dagger a^\dagger \dots \sqrt{1}|1\rangle = a^\dagger \sqrt{n-1}\sqrt{n-2} \dots \sqrt{1}|n-1\rangle = \sqrt{n!}|n\rangle$$

Now, let us consider the Bosons model

1. Analogy with harmonic oscillator model, the creation and annihilation operator can be written as $a_i^\dagger$ and a_i

The i indicate the i -th *single particle state* ψ_i , which has the energy spectrum ω_i . In Bosons system, there can be many i .

For Bosons systems, which are assumed with many single particle states $|n_1 n_2 \dots\rangle$. The system can be written as

$$|n_1 n_2 \dots\rangle = \frac{1}{\sqrt{\prod_i n_i!}} (a_1^\dagger)^{n_1} (a_2^\dagger)^{n_2} \dots |0_1 0_2 \dots\rangle$$

2. We have eigen-equation of $a_i^\dagger$ and a_i like :

$$a_i^\dagger |n_1 n_2 \dots n_i \dots\rangle = \sqrt{n_i + 1} |n_1 n_2 \dots n_i + 1 \dots\rangle; \quad \langle \dots n_i \dots n_2 n_1 | a_i = \sqrt{n_i + 1} \langle \dots n_i + 1 \dots n_2 n_1 |$$

$$a_i |n_1 n_2 \dots n_i \dots\rangle = \sqrt{n_i} |n_1 n_2 \dots n_i - 1 \dots\rangle; \quad \langle \dots n_i \dots n_2 n_1 | a_i^\dagger = \sqrt{n_i} \langle \dots n_i - 1 \dots n_2 n_1 |$$

3. The commutation relation can be read as

$$[a_i, a_i^\dagger] = a_i a_i^\dagger - a_i^\dagger a_i = 1$$

$$[a_i, a_j^\dagger] = a_i a_j^\dagger - a_j^\dagger a_i = 0$$

$$\langle \dots a_i \dots | a_i a_i^\dagger - a_i^\dagger a_i | \dots a_i \dots \rangle = \langle \dots a_i \dots | a_i a_i^\dagger - a_i^\dagger a_i | \dots a_i \dots \rangle$$

$$= \sqrt{a_i + 1} \sqrt{a_i + 1} \langle \dots a_i \dots | \dots a_i \dots \rangle - \sqrt{a_i} \sqrt{a_i} \langle \dots a_i \dots | \dots a_i \dots \rangle$$

$$= a_i + 1 - a_i = 1 \quad (i = j)$$

So, we can get

$$[a_i, a_j^\dagger] = \delta_{ij}$$

$$\langle \dots a_j a_i \dots | a_i a_j^\dagger - a_j^\dagger a_i | \dots a_i a_j \dots \rangle$$

$$= \langle \dots a_j a_i \dots | a_i a_j^\dagger | \dots a_i a_j \dots \rangle - \langle \dots a_j a_i \dots | a_j^\dagger a_i | \dots a_i a_j \dots \rangle$$

$$= \sqrt{a_i} \sqrt{a_j + 1} \langle \dots a_j a_i \dots | \dots (a_i - 1)(a_j + 1) \dots \rangle$$

$$- \sqrt{a_j + 1} \sqrt{a_i} \langle \dots a_j a_i \dots | \dots (a_i - 1)(a_j + 1) \dots \rangle = 0 \quad (i \neq j)$$

We can get

$$[a_i, a_j^\dagger] = \delta_{ij}$$

That means $i \neq j$, the Bosons stratify the Exchange Symmetry;

$i = j$, the Bosons demonstrate the quantum property.

Analogous we can also get with condition $i = j$ as well as $i \neq j$:

$$[a_i, a_j] = [a_i^\dagger, a_j^\dagger] = 0$$

$$\langle \dots a_j a_i \dots | \dots (a_i + 1)(a_j + 1) \dots \rangle = 0$$

$$\langle \dots a_i \dots | \dots (a_i + 2) \dots \rangle = 0$$

That means $i \neq j$ and $i = j$, the Bosons stratify the Exchange Symmetry;

The fermions system different from the bosons, is that commutative anti-symmetry relation.



This anti-symmetry relation will lead to a fact that single particle state α can only occupy by one fermions.

Let us start to study the commutation relation of Fermion from this two fact.

1. The state of Fermion system can be read as $|1_\alpha 1_\beta 1_\gamma \dots\rangle = a_\alpha^\dagger a_\beta^\dagger a_\gamma^\dagger \dots |0_\alpha 0_\beta 0_\gamma \dots\rangle$

Due to the *commutative anti-symmetry relation*, we have $|1_\alpha 1_\beta 1_\gamma \dots\rangle = -|1_\beta 1_\alpha 1_\gamma \dots\rangle$

$$a_\alpha^\dagger a_\beta^\dagger a_\gamma^\dagger \dots |0_\alpha 0_\beta 0_\gamma \dots\rangle = -a_\beta^\dagger a_\alpha^\dagger a_\gamma^\dagger \dots |0_\beta 0_\alpha 0_\gamma \dots\rangle \longrightarrow (a_\alpha^\dagger a_\beta^\dagger + a_\beta^\dagger a_\alpha^\dagger) |1_\gamma \dots\rangle = 0$$

Therefore, we can get the relation

$$\{a_\alpha^\dagger, a_\beta^\dagger\} = 0$$

Now, we show why obtain the single particle state can only occupy one Fermion.

Pauli theory from the commutation anti-symmetry

According to the $\alpha \neq \beta$ we can get $a_\alpha^\dagger a_\alpha^\dagger a_\gamma^\dagger \dots |1_\alpha 1_\alpha 1_\gamma \dots\rangle = -a_\alpha^\dagger a_\alpha^\dagger a_\gamma^\dagger \dots |1_\alpha 1_\alpha 1_\gamma \dots\rangle$ with condition $\alpha = \beta$

$$a_\alpha^\dagger a_\alpha^\dagger a_\gamma^\dagger \dots |0_\alpha 0_\alpha 0_\gamma \dots\rangle + a_\alpha^\dagger a_\alpha^\dagger a_\gamma^\dagger \dots |0_\alpha 0_\alpha 0_\gamma \dots\rangle = 0 \longrightarrow 2a_\alpha^\dagger a_\alpha^\dagger |1_\gamma \dots\rangle = 0 \longrightarrow a_\alpha^\dagger a_\alpha^\dagger = 0$$

$a_\alpha^\dagger a_\alpha^\dagger = 0$ means, two Fermions can not occupy the same particle state.

1. Then we can get commutation relation of the same annihilation operator (a and a).

That need the adjoint condition like $\langle \dots 1_\gamma 1_\beta 1_\alpha | = \langle 0 | \dots a_\gamma a_\beta a_\alpha$

The relation of annihilation can be obtained by $\langle \dots 1_\gamma 1_\beta 1_\alpha | = -\langle \dots 1_\gamma 1_\alpha 1_\beta | \longrightarrow \langle \dots 0_\gamma 0_\beta 0_\alpha | a_\gamma a_\beta a_\alpha = -\langle \dots 0_\gamma 0_\alpha 0_\beta | a_\gamma a_\alpha a_\beta$

$$\langle \dots 1_\gamma | (a_\beta a_\alpha + a_\alpha a_\beta) = 0 \longrightarrow a_\beta a_\alpha + a_\alpha a_\beta = 0 \longrightarrow \{a_\alpha, a_\beta\} = 0$$

Due to the Pauli theory, we can also get $\{a_\alpha, a_\alpha\} = 0$ with condition $\alpha = \beta$.

Now, we get the commutation relation as

$$\{a_{\alpha}^{\dagger}, a_{\beta}^{\dagger}\} = \{a_{\alpha}, a_{\beta}\} = 0$$

That means $\alpha \neq \beta$, the Fermions stratify the Exchange Anti-Symmetry;

$\alpha = \beta$, the Fermions stratify Pauli theory.

Next, we consider the commutation relation between creation $a^{\dagger}$ and annihilation a operator.

There need a *track* that we construct this $a_{\beta}|1_{\alpha}1_{\beta} \dots\rangle$ to calculate $\{a_{\alpha}^{\dagger}, a_{\beta}\}$ with condition $\alpha \neq \beta$ $a_{\alpha}a_{\alpha}^{\dagger}|0_{\alpha}\rangle = a_{\alpha}|1_{\alpha}\rangle = |0_{\alpha}\rangle$

$$a_{\beta}a_{\alpha}^{\dagger}a_{\beta}^{\dagger}a_{\gamma}^{\dagger} \dots |0_{\alpha}0_{\beta}0_{\gamma} \dots\rangle = -a_{\beta}a_{\beta}^{\dagger}a_{\alpha}^{\dagger}a_{\gamma}^{\dagger} \dots |0_{\alpha}0_{\beta}0_{\gamma} \dots\rangle = -a_{\alpha}^{\dagger}a_{\gamma}^{\dagger} \dots |0_{\alpha}0_{\beta}0_{\gamma} \dots\rangle = -a_{\alpha}^{\dagger}(a_{\beta}a_{\beta}^{\dagger})a_{\gamma}^{\dagger} \dots |0_{\alpha}0_{\beta}0_{\gamma} \dots\rangle$$

$$\longrightarrow a_{\beta}a_{\alpha}^{\dagger}a_{\beta}^{\dagger}a_{\gamma}^{\dagger} \dots |0_{\alpha}0_{\beta}0_{\gamma} \dots\rangle = -a_{\alpha}^{\dagger}a_{\beta}a_{\beta}^{\dagger}a_{\gamma}^{\dagger} \dots |0_{\alpha}0_{\beta}0_{\gamma} \dots\rangle \longrightarrow (a_{\beta}a_{\alpha}^{\dagger} + a_{\alpha}^{\dagger}a_{\beta})|1_{\beta}1_{\gamma} \dots\rangle = 0$$

$$a_{\alpha}^{\dagger}a_{\beta} + a_{\beta}a_{\alpha}^{\dagger} = 0 \longrightarrow \{a_{\alpha}^{\dagger}, a_{\beta}\} = 0$$

If $\alpha = \beta$, we can add the $a_\alpha a_\alpha^\dagger$ and $a_\alpha^\dagger a_\alpha$ in $|1_\alpha 1_\beta \dots\rangle$ to obtain

$$a_\alpha a_\alpha^\dagger a_\alpha^\dagger a_\alpha^\dagger a_\gamma^\dagger \dots |0_\alpha 0_\beta 0_\gamma \dots\rangle = a_\alpha a_\alpha^\dagger |1_\alpha 1_\beta 1_\gamma \dots\rangle = 0$$

$$a_\alpha^\dagger a_\alpha a_\alpha^\dagger a_\beta^\dagger a_\gamma^\dagger \dots |0_\alpha 0_\beta 0_\gamma \dots\rangle = a_\alpha^\dagger a_\alpha |1_\alpha 1_\beta 1_\gamma \dots\rangle = |1_\alpha 1_\beta 1_\gamma \dots\rangle$$

$$(a_\alpha^\dagger a_\alpha + a_\alpha a_\alpha^\dagger) |1_\alpha 1_\beta 1_\gamma \dots\rangle = |1_\alpha 1_\beta 1_\gamma \dots\rangle + 0$$

$$\longrightarrow (a_\alpha^\dagger a_\alpha + a_\alpha a_\alpha^\dagger) = 1 \longrightarrow \{a_\alpha^\dagger, a_\alpha\} = 1$$

So, we get the commutation relation as

$$\{a_\alpha, a_\beta^\dagger\} = \delta_{\alpha\beta}$$

That means $\alpha \neq \beta$, the Fermion stratify the Exchange Anti-Symmetry;

$\alpha = \beta$, the Fermion demonstrate the quantum property.

Results:

1. For Bosons systems

$$[a_i, a_j^\dagger] = \delta_{ij}, \quad [a_i, a_j] = [a_i^\dagger, a_j^\dagger] = 0$$

2. For Fermion systems

$$\{a_\alpha, a_\beta^\dagger\} = \delta_{\alpha\beta}, \quad \{a_\alpha, a_\beta\} = \{a_\alpha^\dagger, a_\beta^\dagger\} = 0$$

1. For a spin system, there are three basis operator, i.e, the Pauli operator

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$S_\alpha = \frac{1}{2} \sigma_\alpha, \text{ where } \alpha = x, y, z.$$

2. Analysis to *creation and annihilation* of Bosons and Fermion, we construct the Spin *Plus* and *Minus* operator like

$$S^+ = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad S^- = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \quad \begin{aligned} S^+ &= (\sigma_x - i\sigma_y) / 2 \\ S^- &= (\sigma_x + i\sigma_y) / 2 \end{aligned}$$

3. There we have the relation between S^+ , S^- and S_z :

$$\{S^+, S_z\} = \{S^-, S_z\} = 0$$

Analysis to the *process* of Bosons and Fermion, we calculate the commutation relation of S^+ and S^- .

1. For $i = j$:

$$\{S_i, S_i^\dagger\} = S_i S_i^\dagger + S_i^\dagger S_i = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = I$$

$$\{S_i^\dagger, S_i^\dagger\} = S_i^\dagger S_i^\dagger + S_i^\dagger S_i^\dagger = 0$$

$$\{S_i, S_i\} = S_i S_i + S_i S_i = 0$$

- This commutation relation satisfying the Fermion.
- That means for the same site, the spins are like the Fermion.

2. For $i \neq j$:

$$\begin{aligned} [S_i, S_j^\dagger] &= S_i S_j^\dagger - S_j^\dagger S_i = (S_i \otimes I_j)(I_i \otimes S_j^\dagger) - (I_i \otimes S_j^\dagger)(S_i \otimes I_j) \\ &= S_i \otimes S_j^\dagger - S_i \otimes S_j^\dagger = 0 \end{aligned}$$

$$\begin{aligned} [S_i^\dagger, S_j^\dagger] &= S_i^\dagger S_j^\dagger - S_i^\dagger S_j^\dagger = (S_i^\dagger \otimes I_j)(I_i \otimes S_j^\dagger) - (I_i \otimes S_j^\dagger)(S_i^\dagger \otimes I_j) \\ &= S_i^\dagger \otimes S_j^\dagger - S_i^\dagger \otimes S_j^\dagger = 0 \end{aligned}$$

$$\begin{aligned} [S_i, S_j] &= S_i S_j - S_i S_j = (S_i \otimes I_j)(I_i \otimes S_j) - (I_i \otimes S_j)(S_i \otimes I_j) \\ &= S_i \otimes S_j - S_i \otimes S_j = 0 \end{aligned}$$

- This commutation relation satisfying the Bosons.
- But it need to consider the different particle state $i \neq j$.
- To this case, in Spin system we call that the *Hard-Core* Bosons.

Results:

1. For $i = j$

$$\{S_i, S_i^\dagger\} = 1, \quad \{S_i, S_i\} = \{S_i^\dagger, S_i^\dagger\} = 0$$

1. For $i \neq j$

$$[S_i, S_j^\dagger] = 0, \quad [S_i, S_j] = [S_i^\dagger, S_j^\dagger] = 0$$

The JW translation can allow to solve the Bosons system by mapping it to non-interacting Fermions.

1. First, we introduce this JW translation relation

$$S^+ = F_i c_i^+, \quad S^- = F_i c_i,$$

$$F_i = \prod_{n=1}^{i-1} \sigma_n^z = \prod_{n=1}^{i-1} (2c_n^\dagger c_n - 1) \quad \text{The Jordan-Wigner strings}$$

$$\sigma_i^z = 2c_i^\dagger c_i - 1, \quad \sigma_i^x = F_i (c_i + c_i^\dagger), \quad \sigma_i^y = iF_i (c_i - c_i^\dagger),$$

Single site system

$$S^+ = (\sigma_x - i\sigma_y) / 2$$

$$S = (\sigma_x + i\sigma_y) / 2$$

2. Then, we check this translation of Fermion c_i $c_i^\dagger$ represent by **Spins (Hard Core Bosons)** S, S^+ with JW strings.

$$\{c_i, c_j^\dagger\} = \delta_{ij}, \quad \{c_i, c_j\} = \{c_i^\dagger, c_j^\dagger\} = 0$$

For $i = j$

$$\begin{aligned}
 \{S_i, S_i^\dagger\} &= S_i S_i^\dagger + S_i^\dagger S_i = F_i c_i^- F_i c_i^+ + F_i c_i^+ F_i c_i^- = \sigma_1^Z \sigma_2^Z \dots \sigma_{i-1}^Z c_i^- \sigma_1^Z \sigma_2^Z \dots \sigma_{i-1}^Z c_i^+ + \sigma_1^Z \sigma_2^Z \dots \sigma_{i-1}^Z c_i^+ \sigma_1^Z \sigma_2^Z \dots \sigma_{i-1}^Z c_i^- \\
 &= \sigma_1^{Z^2} \sigma_2^{Z^2} \dots \sigma_{i-1}^{Z^2} c_i^- c_i^+ + \sigma_1^{Z^2} \sigma_2^{Z^2} \dots \sigma_{i-1}^{Z^2} c_i^+ c_i^- \quad c_i^- \sigma_{j \neq i}^Z - \sigma_{j \neq i}^Z c_i^- = 0 \\
 &\quad c_i^+ \sigma_{j \neq i}^Z - \sigma_{j \neq i}^Z c_i^+ = 0 \\
 \{c_i, c_i^\dagger\} = 1 \quad \longleftarrow &= c_i^- c_i^+ + c_i^+ c_i^- = 1
 \end{aligned}$$

$$\begin{aligned}
 \{S_i, S_i\} &= S_i S_i + S_i S_i = F_i c_i^- F_i c_i^- + F_i c_i^- F_i c_i^- = \sigma_1^Z \sigma_2^Z \dots \sigma_{i-1}^Z c_i^- \sigma_1^Z \sigma_2^Z \dots \sigma_{i-1}^Z c_i^- + \sigma_1^Z \sigma_2^Z \dots \sigma_{i-1}^Z c_i^- \sigma_1^Z \sigma_2^Z \dots \sigma_{i-1}^Z c_i^- \\
 &= \sigma_1^{Z^2} \sigma_2^{Z^2} \dots \sigma_{i-1}^{Z^2} c_i^- c_i^- + \sigma_1^{Z^2} \sigma_2^{Z^2} \dots \sigma_{i-1}^{Z^2} c_i^- c_i^- \\
 \{c_i, c_i\} = 0 \quad \longleftarrow &= c_i^- c_i^- + c_i^- c_i^- = 0
 \end{aligned}$$

$$\begin{aligned}
 \{S_i^\dagger, S_i^\dagger\} &= S_i^\dagger S_i^\dagger + S_i^\dagger S_i^\dagger = F_i c_i^+ F_i c_i^+ + F_i c_i^+ F_i c_i^+ = \sigma_1^Z \sigma_2^Z \dots \sigma_{i-1}^Z c_i^+ \sigma_1^Z \sigma_2^Z \dots \sigma_{i-1}^Z c_i^+ + \sigma_1^Z \sigma_2^Z \dots \sigma_{i-1}^Z c_i^+ \sigma_1^Z \sigma_2^Z \dots \sigma_{i-1}^Z c_i^+ \\
 &= \sigma_1^{Z^2} \sigma_2^{Z^2} \dots \sigma_{i-1}^{Z^2} c_i^+ c_i^+ + \sigma_1^{Z^2} \sigma_2^{Z^2} \dots \sigma_{i-1}^{Z^2} c_i^+ c_i^+ \\
 \{c_i^\dagger, c_i^\dagger\} = 0 \quad \longleftarrow &= c_i^+ c_i^+ + c_i^+ c_i^+ = 0
 \end{aligned}$$

$$\longrightarrow \{c_i, c_i^\dagger\} = 1, \quad \{c_i, c_i\} = \{c_i^\dagger, c_i^\dagger\} = 0$$

For $i \neq j$

Single site system $c_i^+ = S_i^+, c_i^- = S_i^-$

$$[S_i, S_j^\dagger] = S_i S_j^\dagger - S_j^\dagger S_i = F_i c_i^- F_j c_j^+ - F_j c_j^+ F_i c_i^- = \sigma_1^Z \sigma_2^Z \dots \sigma_{i-1}^Z c_i^- \sigma_1^Z \sigma_2^Z \dots \sigma_i^Z \dots \sigma_{j-1}^Z c_j^+ - \sigma_1^Z \sigma_2^Z \dots \sigma_i^Z \dots \sigma_{j-1}^Z c_j^+ \sigma_1^Z \sigma_2^Z \dots \sigma_{i-1}^Z c_i^-$$

$$\{c_i, c_j^\dagger\} = 0 \quad \longleftarrow \quad = c_i^- \sigma_i^Z \sigma_{i+1}^Z \dots \sigma_j^Z c_j^+ + \sigma_i^Z c_i^- \sigma_{i+1}^Z \dots \sigma_j^Z c_j^+ = 0$$

$$\{c^-, \sigma_i^Z\} = c_i^- \sigma_i^Z + \sigma_i^Z c_i^- = 0$$

Understand with MPO

$$[S_i, S_j] = S_i S_j - S_j S_i = F_i S_i^- F_j S_j^- + F_j S_j^- F_i S_i^- = \sigma_1^Z \sigma_2^Z \dots \sigma_{i-1}^Z c_i^- \sigma_1^Z \sigma_2^Z \dots \sigma_i^Z \dots \sigma_{j-1}^Z c_j^- - \sigma_1^Z \sigma_2^Z \dots \sigma_i^Z \dots \sigma_{j-1}^Z c_j^- \sigma_1^Z \sigma_2^Z \dots \sigma_{i-1}^Z c_i^-$$

$$\{c_i, c_j\} = 0 \quad \longleftarrow \quad = c_i^- \sigma_i^Z \sigma_{i+1}^Z \dots \sigma_j^Z c_j^- + \sigma_i^Z c_i^- \sigma_{i+1}^Z \dots \sigma_j^Z c_j^- = 0$$

$$\{c^-, \sigma_i^Z\} = c_i^- \sigma_i^Z + \sigma_i^Z c_i^- = 0$$

$$[S_i^\dagger, S_i^\dagger] = S_i^\dagger S_i^\dagger + S_i^\dagger S_i^\dagger = F_i S_i^+ F_i S_i^+ + F_i S_i^+ F_i S_i^+ = \sigma_1^Z \sigma_2^Z \dots \sigma_{i-1}^Z c_i^+ \sigma_1^Z \sigma_2^Z \dots \sigma_i^Z \dots \sigma_{j-1}^Z c_j^+ - \sigma_1^Z \sigma_2^Z \dots \sigma_i^Z \dots \sigma_{j-1}^Z c_j^+ \sigma_1^Z \sigma_2^Z \dots \sigma_{i-1}^Z c_i^+$$

$$\{c_i^\dagger, c_j^\dagger\} = 0 \quad \longleftarrow \quad = c_i^+ \sigma_i^Z \sigma_{i+1}^Z \dots \sigma_j^Z c_j^+ + \sigma_i^Z c_i^+ \sigma_{i+1}^Z \dots \sigma_j^Z c_j^+ = 0$$

$$\{c^+, \sigma_i^Z\} = c_i^+ \sigma_i^Z + \sigma_i^Z c_i^+ = 0$$

$$\longrightarrow \quad \{c_i, c_j^\dagger\} = 0, \quad \{c_i, c_j\} = \{c_i^\dagger, c_j^\dagger\} = 0$$

Results:

$$S_i^+ = F_i c_i^+, \quad S_i^- = F_i c_i^-,$$

$$i = j$$

$$\{S_i^-, S_i^+\} = 1,$$

$$\{S_i^-, S_i^-\} = \{S_i^+, S_i^+\} = 0$$

$$i \neq j$$

$$[S_i^-, S_i^+] = 0$$

$$[S_i^-, S_i^-] = [S_i^+, S_i^+] = 0$$

$$\{c_i^-, c_j^+\} = \delta_{ij},$$

$$\{c_i^-, c_j^-\} = \{c_i^+, c_j^+\} = 0$$

$$S^+ = (\sigma_x - i\sigma_y) / 2$$

$$S = (\sigma_x + i\sigma_y) / 2$$

$$\begin{cases} \sigma_i^z = 2c_i^\dagger c_i - 1, \\ \sigma_i^x = F_i(c_i + c_i^\dagger), \\ \sigma_i^y = iF_i(c_i - c_i^\dagger), \end{cases}$$

So, the Spin operator has been provided that it can represent by non-interaction Fermions with JW string F_i

We solve analysis the Bosons system with JW translation.

1. Consider the XY model with external field along Z direction, whose Hamiltonian reads

$$H = \sum_i [(\lambda + \gamma)\sigma_i^x \sigma_{i+1}^x + (\lambda - \gamma)\sigma_i^y \sigma_{i+1}^y] + h_z \sum_i \sigma_i^z$$

2. Represent it by Fermion operator with JW translation $\sigma_i^z = 2c_i^\dagger c_i - 1$, $\sigma_i^x = F_i(c_i + c_i^\dagger)$, $\sigma_i^y = iF_i(c_i - c_i^\dagger)$,

$$H = \sum_i [(\lambda + \gamma)\sigma_i^x \sigma_{i+1}^x + (\lambda - \gamma)\sigma_i^y \sigma_{i+1}^y] + h_z \sum_i \sigma_i^z$$



$$H_{JW} = \sum_i [\lambda(c_i c_{i+1}^\dagger - c_i^\dagger c_{i+1}) + \gamma(c_i c_{i+1} - c_i^\dagger c_{i+1}^\dagger)] + h_z \sum_i (c_i^\dagger c_i - c_i c_i^\dagger)$$

$$\{c_i, c_i^\dagger\} = c_i c_i^\dagger + c_i^\dagger c_i = 1$$

Fermion system is

PBC $i = 1 = N + 1$

$$(\lambda + \gamma)\sigma_i^x \sigma_{i+1}^x + (\lambda - \gamma)\sigma_i^y \sigma_{i+1}^y$$

$$= [(\lambda + \gamma)(c_i + c_i^\dagger)\sigma_i^z(c_{i+1} + c_{i+1}^\dagger)] + [-(\lambda - \gamma)(c_i - c_i^\dagger)\sigma_i^z(c_{i+1} - c_{i+1}^\dagger)]$$



$$= [(\lambda + \gamma)(c_i - c_i^\dagger)(c_{i+1} + c_{i+1}^\dagger)] + [-(\lambda - \gamma)(c_i + c_i^\dagger)(c_{i+1} - c_{i+1}^\dagger)]$$



$$= (\lambda + \gamma)(c_i c_{i+1} + c_i c_{i+1}^\dagger - c_i^\dagger c_{i+1} - c_i^\dagger c_{i+1}^\dagger) + (\lambda - \gamma)(-c_i c_{i+1} + c_i c_{i+1}^\dagger - c_i^\dagger c_{i+1} + c_i^\dagger c_{i+1}^\dagger)$$

$$c_i \sigma_i^z = c_i, \\ c_i^\dagger \sigma_i^z = -c_i^\dagger$$

Fourier transformation for H_{JW}

1. We apply the FT for $c_{i(n)}$ and $c_{i+1(n+1)}^\dagger$.

The inverse transformation of f_k and $f_k^\dagger$

$$f_k = \frac{1}{\sqrt{N}} \sum_n e^{-ikn} c_n, \quad f_k^\dagger = \frac{1}{\sqrt{N}} \sum_n e^{-ikn} c_n^\dagger$$

$$c_n = \frac{1}{\sqrt{N}} \sum_k e^{-ink} f_k, \quad c_n^\dagger = \frac{1}{\sqrt{N}} \sum_k e^{ink} f_k^\dagger$$

where k is in the Brillouin zone BZ with $k = -\pi + 2\pi n/N$ with $n \in [0, N-1]$

2. Then we can obtain the H_{JW} with f_k and $f_k^\dagger$.

Hopping term

$$\begin{aligned} \sum_n (c_n c_{n+1}^\dagger - c_n^\dagger c_{n+1}) &= \sum_n \left(\frac{1}{\sqrt{N}} \sum_k e^{-ink} f_k \cdot \frac{1}{\sqrt{N}} \sum_{k'} e^{i(n+1)k'} f_{k'}^\dagger - \frac{1}{\sqrt{N}} \sum_k e^{ink} f_k^\dagger \cdot \frac{1}{\sqrt{N}} \sum_{k'} e^{-i(n+1)k'} f_{k'} \right) \\ &= \sum_{kk'} e^{ik' \delta_{kk'}} N f_k f_{k'}^\dagger - e^{-ik' \delta_{kk'}} N f_k^\dagger f_{k'} = \sum_k e^{ik} f_k f_k^\dagger - e^{-ik} f_k^\dagger f_k \\ &= \sum_{k>0} e^{-ik} f_{-k} f_{-k}^\dagger + \sum_{k>0} e^{ik} f_k f_k^\dagger - \sum_{k>0} e^{ik} f_{-k}^\dagger f_{-k} - \sum_{k>0} e^{-ik} f_k^\dagger f_k \end{aligned}$$

$$= \sum_{k>0} e^{-ik} f_{-k} f_{-k}^{\dagger} + \sum_{k>0} e^{ik} f_k f_k^{\dagger} - \sum_{k>0} e^{ik} f_{-k}^{\dagger} f_{-k} - \sum_{k>0} e^{-ik} f_k^{\dagger} f_k$$

$$\{c_i, c_i^{\dagger}\} = c_i c_i^{\dagger} + c_i^{\dagger} c_i = 1$$

$$= \sum_{k>0} e^{-ik} f_{-k} f_{-k}^{\dagger} - e^{ik} (1 - f_{-k} f_{-k}^{\dagger}) + e^{ik} (1 - f_k^{\dagger} f_k) - e^{-ik} f_k^{\dagger} f_k$$

$$\cos k = e^{-ik} + e^{ik}$$

$$= \sum_{k>0} (e^{-ik} + e^{ik}) f_{-k} f_{-k}^{\dagger} - (e^{-ik} + e^{ik}) f_k^{\dagger} f_k = \sum_{k>0} 2 \cos k (f_{-k} f_{-k}^{\dagger} - f_k^{\dagger} f_k)$$

Pairing term

Potential term

$$\sum_n (c_n c_{n+1} - c_n^{\dagger} c_{n+1}^{\dagger}) = \sum_{k>0} 2i \sin k (f_k^{\dagger} f_{-k}^{\dagger} - f_{-k} f_k)$$

$$\sum_n (c_n^{\dagger} c_n - c_n c_n^{\dagger}) = \sum_{k>0} 2 (f_k^{\dagger} f_k - f_{-k} f_{-k}^{\dagger})$$

The H_{JW} can be written as

$$H_{JW} = \sum_{k>0} 2\lambda \cos k (f_{-k} f_{-k}^{\dagger} - f_k^{\dagger} f_k) + 2i\gamma \sin k (f_k^{\dagger} f_{-k}^{\dagger} - f_{-k} f_k) + h_z (f_k^{\dagger} f_k - f_{-k} f_{-k}^{\dagger})$$

3. Finally, we get the Hamiltonian with 2D matrix form

$$\begin{aligned}
 & \sum_{k>0} [f_k^\dagger, f_{-k}] \omega_k \begin{bmatrix} \cos \theta_k & \sin \theta_k e^{-i\pi/2} \\ \sin \theta_k e^{i\pi/2} & -\cos \theta_k \end{bmatrix} \begin{bmatrix} f_k \\ f_{-k}^\dagger \end{bmatrix} \quad \omega_k = 2 \sqrt{\left[\frac{h_z}{2} - \lambda \cos k\right]^2 + [\gamma \sin k]^2} \quad \sin \theta_k = \frac{2\gamma \sin k}{\omega_k} \\
 &= \sum_{k>0} \omega_k (\cos \theta_k f_k^\dagger f_k + \sin \theta_k e^{i\pi/2} f_k f_k + \sin \theta_k e^{-i\pi/2} f_k^\dagger f_{-k}^\dagger - \cos \theta_k f_{-k} f_{-k}^\dagger) \quad \cos \theta_k = \sqrt{1 - \sin^2 \theta_k} \\
 &= \sum_{k>0} (h_z - 2\lambda \cos k) f_k^\dagger f_k + 2\gamma \sin k e^{i\pi/2} f_k f_k + 2\gamma \sin k e^{-i\pi/2} f_k^\dagger f_{-k}^\dagger - (h_z - 2\lambda \cos k) f_{-k} f_{-k}^\dagger = \sqrt{1 - \frac{\gamma^2 \sin^2 k}{\left[\frac{h_z}{2} - \lambda \cos k\right]^2 + \gamma^2 \sin^2 k}} \\
 &= \sum_{k>0} 2\lambda \cos k (f_{-k} f_{-k}^\dagger - f_k^\dagger f_k) + h_z (f_k^\dagger f_k - f_{-k} f_{-k}^\dagger) + 2\gamma \sin k (e^{i\pi/2} f_k f_k + e^{-i\pi/2} f_k^\dagger f_{-k}^\dagger) \\
 &= \sum_{k>0} 2\lambda \cos k (f_{-k} f_{-k}^\dagger - f_k^\dagger f_k) + h_z (f_k^\dagger f_k - f_{-k} f_{-k}^\dagger) + 2\gamma \sin k (i f_k f_k - i f_k^\dagger f_{-k}^\dagger) = \frac{\left[\frac{h_z}{2} - \lambda \cos k\right]^2}{\sqrt{\left[\frac{h_z}{2} - \lambda \cos k\right]^2 + \gamma^2 \sin^2 k}} \\
 &= \sum_{k>0} 2\lambda \cos k (f_{-k} f_{-k}^\dagger - f_k^\dagger f_k) + h_z (f_k^\dagger f_k - f_{-k} f_{-k}^\dagger) + 2\gamma i \sin k (f_k f_k - f_k^\dagger f_{-k}^\dagger) = \frac{h_z - 2\lambda \cos k}{\omega_k} \\
 &H_{JW} = \sum_{k>0} 2\lambda \cos k (f_{-k} f_{-k}^\dagger - f_k^\dagger f_k) + 2i\gamma \sin k (f_k^\dagger f_{-k}^\dagger - f_{-k} f_k) + h_z (f_k^\dagger f_k - f_{-k} f_{-k}^\dagger)
 \end{aligned}$$

$$H_{JW} = \sum_{k>0} \Psi_k^\dagger M_k \Psi_k$$

$$\Psi_k^\dagger = [f_k^\dagger, f_{-k}], \Psi_k = \begin{bmatrix} f_k \\ f_{-k}^\dagger \end{bmatrix} \quad M_k = \omega_k \begin{bmatrix} \cos \theta_k & \sin \theta_k e^{-i\pi/2} \\ \sin \theta_k e^{i\pi/2} & -\cos \theta_k \end{bmatrix} \quad \omega_k = 2 \sqrt{\left[\frac{h_z}{2} - \lambda \cos k \right]^2 + [\gamma \sin k]^2} \quad \sin \theta_k = \frac{2\gamma \sin k}{\omega_k}$$

The follow aim is to diagonalize the H_{JW} , i.e., Find a new basic to cancel the non-diagonal element of M_k

$$H_{JW} = \sum_{k>0} \Psi_k^\dagger M_k \Psi_k = \sum_{k>0} \Psi_k^\dagger U_k^\dagger U_k M_k U_k^\dagger U_k \Psi_k = \sum_{k>0} \Phi_k^\dagger U_k M_k U_k^\dagger \Phi_k$$

$$U_k = \begin{pmatrix} e^{i\pi/4} \cos(\theta_k/2) & e^{-i\pi/4} \sin(\theta_k/2) \\ -e^{i\pi/4} \sin(\theta_k/2) & e^{-i\pi/4} \cos(\theta_k/2) \end{pmatrix}$$

$$U_k^\dagger = \begin{pmatrix} e^{-i\pi/4} \cos(\theta_k/2) & -e^{-i\pi/4} \sin(\theta_k/2) \\ e^{i\pi/4} \sin(\theta_k/2) & e^{i\pi/4} \cos(\theta_k/2) \end{pmatrix}$$

$$= \sum_{k>0} \omega_k \Phi_k^\dagger \begin{pmatrix} e^{i\pi/4} \cos(\theta_k/2) & e^{-i\pi/4} \sin(\theta_k/2) \\ -e^{i\pi/4} \sin(\theta_k/2) & e^{-i\pi/4} \cos(\theta_k/2) \end{pmatrix} \begin{bmatrix} \cos \theta_k & \sin \theta_k e^{-i\pi/2} \\ \sin \theta_k e^{i\pi/2} & -\cos \theta_k \end{bmatrix} \begin{pmatrix} e^{-i\pi/4} \cos(\theta_k/2) & -e^{-i\pi/4} \sin(\theta_k/2) \\ e^{i\pi/4} \sin(\theta_k/2) & e^{i\pi/4} \cos(\theta_k/2) \end{pmatrix} \Phi_k$$

$$= \sum_{k>0} \omega_k \Phi_k^\dagger \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \Phi_k$$

We call this the **Bogo Liubov** transformation $\Phi_k^\dagger = \Psi_k^\dagger U_k^\dagger \equiv [c_k^\dagger, c_{-k}]$, $\Phi_k = U_k \Psi_k \equiv \begin{bmatrix} c_k \\ c_{-k}^\dagger \end{bmatrix}$

1. We can obtain the diagonal Hamiltonian H_{JW} with basic $\Phi_k (c_k)$

$$\begin{aligned}
 H_{JW} &= \sum_{k>0} \omega_k \Phi_k^\dagger \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \Phi_k = \sum_{k>0} \omega_k [c_k^\dagger, c_{-k}] \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} c_k \\ c_{-k}^\dagger \end{bmatrix} = \sum_{k>0} \omega_k (c_k^\dagger c_k - c_{-k} c_{-k}^\dagger) \\
 &= \sum_k \omega_k \left(c_k^\dagger c_k - \frac{1}{2} \right) \quad \text{Energy term } \omega_k = 2\sqrt{\left[\frac{\hbar_z}{2} - \lambda \cos k\right]^2 + [\gamma \sin k]^2} \\
 &= \sum_{k>0} \omega_k (c_k^\dagger c_k + c_{-k}^\dagger c_{-k} - 1) \\
 &= \sum_k \frac{1}{2} \omega_k (c_k^\dagger c_k + c_k^\dagger c_k - 1) \\
 &= \sum_k \frac{1}{2} \omega_k \left(c_k^\dagger c_k - \frac{1}{2} \right)
 \end{aligned}$$

2. In Heisenberg picture, we get the time evolution of diagonal Basic Φ_k

$$\begin{aligned}
 \Phi_k(t) &= e^{iHt} \Phi_k(0) e^{-iHt} \\
 &= e^{\sum_{k>0} \omega_k (c_k^\dagger c_k - c_{-k} c_{-k}^\dagger)t} \begin{bmatrix} c_k(0) \\ c_{-k}^\dagger(0) \end{bmatrix} e^{-\sum_{k>0} \omega_k (c_k^\dagger c_k - c_{-k} c_{-k}^\dagger)t} = \begin{bmatrix} e^{\sum_{k>0} \omega_k (c_k^\dagger c_k - c_{-k} c_{-k}^\dagger)t} c_k(0) e^{-\sum_{k>0} \omega_k (c_k^\dagger c_k - c_{-k} c_{-k}^\dagger)t} \\ e^{\sum_{k>0} \omega_k (c_k^\dagger c_k - c_{-k} c_{-k}^\dagger)t} c_{-k}^\dagger(0) e^{-\sum_{k>0} \omega_k (c_k^\dagger c_k - c_{-k} c_{-k}^\dagger)t} \end{bmatrix} \\
 &= \begin{bmatrix} e^{\sum_{k>0} \omega_k (c_k^\dagger c_k - c_{-k} c_{-k}^\dagger)t} c_k(0) e^{-\sum_{k>0} \omega_k (c_k^\dagger c_k - c_{-k} c_{-k}^\dagger)t} \\ e^{\sum_{k>0} \omega_k (c_k^\dagger c_k - c_{-k} c_{-k}^\dagger)t} c_{-k}^\dagger(0) e^{-\sum_{k>0} \omega_k (c_k^\dagger c_k - c_{-k} c_{-k}^\dagger)t} \end{bmatrix} = \begin{bmatrix} e^{-i\omega_k t} c_k(0) \\ e^{i\omega_k t} c_{-k}^\dagger(0) \end{bmatrix} = \begin{pmatrix} e^{-i\omega_k t} & 0 \\ 0 & e^{i\omega_k t} \end{pmatrix} \begin{bmatrix} c_k(0) \\ c_{-k}^\dagger(0) \end{bmatrix}
 \end{aligned}$$

$$e^{\hat{A}} \hat{B} e^{\hat{A}} = \sum_i \frac{1}{i!} [\hat{A}^i, \hat{B}]$$

$$\Phi_k(t) = \begin{pmatrix} e^{-i\omega_k t} & 0 \\ 0 & e^{i\omega_k t} \end{pmatrix} \Phi_k(0)$$

3. Then transform back to $f_k(t)$

$$\Psi_k(t) = U_k^\dagger \Phi_k(t) = \begin{pmatrix} e^{-i\pi/4} \cos(\theta_k/2) & -e^{-i\pi/4} \sin(\theta_k/2) \\ e^{i\pi/4} \sin(\theta_k/2) & e^{i\pi/4} \cos(\theta_k/2) \end{pmatrix} \begin{bmatrix} e^{-i\omega_k t} c_k(0) \\ e^{i\omega_k t} c_{-k}^\dagger(0) \end{bmatrix}$$

$$\begin{bmatrix} f_k(t) \\ f_{-k}^\dagger(t) \end{bmatrix} = \begin{bmatrix} e^{-i\pi/4} e^{-i\omega_k t} \cos(\theta_k/2) c_k(0) - e^{-i\pi/4} e^{i\omega_k t} \sin(\theta_k/2) c_{-k}^\dagger(0) \\ e^{i\pi/4} e^{-i\omega_k t} \sin(\theta_k/2) c_k(0) + e^{i\pi/4} e^{i\omega_k t} \cos(\theta_k/2) c_{-k}^\dagger(0) \end{bmatrix}$$

$$f_k(t) = e^{-i\pi/4} e^{-i\omega_k t} \cos(\theta_k/2) c_k(0) - e^{-i\pi/4} e^{i\omega_k t} \sin(\theta_k/2) c_{-k}^\dagger(0)$$

$$f_k^\dagger(t) = e^{i\pi/4} e^{i\omega_k t} \cos(\theta_k/2) c_k^\dagger(0) - e^{i\pi/4} e^{-i\omega_k t} \sin(\theta_k/2) c_{-k}(0)$$

$$f_{-k}^\dagger(t) = e^{i\pi/4} e^{-i\omega_k t} \sin(\theta_k/2) c_k(0) + e^{i\pi/4} e^{i\omega_k t} \cos(\theta_k/2) c_{-k}^\dagger(0)$$

$$f_{-k}(t) = e^{-i\pi/4} e^{i\omega_k t} \sin(\theta_k/2) c_k^\dagger(0) + e^{-i\pi/4} e^{-i\omega_k t} \cos(\theta_k/2) c_{-k}(0)$$

4. Then transform back to $f_k(0)$

$$\begin{aligned} c_k(0) &= e^{i\pi/4} \cos(\theta_k/2) f_k(0) + e^{-i\pi/4} \sin(\theta_k/2) f_{-k}^\dagger(0) & c_{-k}^\dagger(0) &= -e^{i\pi/4} \sin(\theta_k/2) f_k(0) + e^{-i\pi/4} \cos(\theta_k/2) f_{-k}^\dagger(0) \\ c_k^\dagger(0) &= e^{-i\pi/4} \cos(\theta_k/2) f_k^\dagger(0) + e^{i\pi/4} \sin(\theta_k/2) f_{-k}(0) & c_{-k}(0) &= -e^{-i\pi/4} \sin(\theta_k/2) f_k^\dagger(0) + e^{i\pi/4} \cos(\theta_k/2) f_{-k}(0) \end{aligned}$$

$$\begin{aligned} f_k(t) &= (\cos^2(\theta_k/2) e^{-i\omega_k t} + \sin^2(\theta_k/2) e^{i\omega_k t}) f_k(0) + (e^{-i\omega_k t} - e^{i\omega_k t}) e^{-i\pi/2} \sin(\theta_k/2) \cos(\theta_k/2) f_{-k}^\dagger(0) \\ f_k^\dagger(t) &= (\cos^2(\theta_k/2) e^{i\omega_k t} + \sin^2(\theta_k/2) e^{-i\omega_k t}) f_k^\dagger(0) + (e^{i\omega_k t} - e^{-i\omega_k t}) e^{i\pi/2} \sin(\theta_k/2) \cos(\theta_k/2) f_{-k}(0) \\ f_{-k}^\dagger(t) &= (e^{-i\omega_k t} - e^{i\omega_k t}) e^{i\pi/2} \sin(\theta_k/2) \cos(\theta_k/2) f_k(0) + (e^{-i\omega_k t} \sin^2(\theta_k/2) + e^{i\omega_k t} \cos^2(\theta_k/2)) f_{-k}^\dagger(0) \\ f_{-k}(t) &= (e^{i\omega_k t} - e^{-i\omega_k t}) e^{-i\pi/2} \sin(\theta_k/2) \cos(\theta_k/2) f_k^\dagger(0) + (e^{i\omega_k t} \sin^2(\theta_k/2) + e^{-i\omega_k t} \cos^2(\theta_k/2)) f_{-k}(0) \end{aligned}$$

5. The result we get the relation of $f_k(t)$ with $f_k(0)$

$$\sigma_\alpha - \text{JW} \rightarrow f_i(0) - \text{FT} \rightarrow f_k(0) - \text{BT} \rightarrow c_k(0) - \text{Hei Pic} \rightarrow c_k(t)$$

$\downarrow \text{Inv BT}$

The non-interaction Fermion presentation of Spin model

$f_k(0) \leftarrow \text{represent} - c_k(0) \leftarrow \text{represent} - f_k(t)$

The observation of H_B

$$H_B = \sum_n c_n^\dagger c_n = \sum_k f_k^\dagger f_k$$

$$\langle H_B(t) \rangle = \left\langle \sum_k \left(f_k^\dagger(t) f_k(t) \right) \right\rangle$$

$$= \left\langle \sum_k \left((\cos^2(\theta_k/2) e^{i\omega_k t} + \sin^2(\theta_k/2) e^{-i\omega_k t}) f_k^\dagger(0) + (e^{i\omega_k t} - e^{-i\omega_k t}) e^{i\pi/2} \sin(\theta_k/2) \cos(\theta_k/2) f_{-k}(0) \right) (\cos^2(\theta_k/2) e^{-i\omega_k t} + \sin^2(\theta_k/2) e^{i\omega_k t}) f_k(0) + (e^{-i\omega_k t} - e^{i\omega_k t}) e^{-i\pi/2} \sin(\theta_k/2) \cos(\theta_k/2) f_{-k}^\dagger(0) \right\rangle$$

$$= \left\langle \sum_k (e^{i\omega_k t} - e^{-i\omega_k t}) \sin(\theta_k/2) \cos(\theta_k/2) f_{-k}(0) (e^{-i\omega_k t} - e^{i\omega_k t}) \sin(\theta_k/2) \cos(\theta_k/2) f_{-k}^\dagger(0) \right\rangle$$

$$= \left\langle \sum_k (e^{i\omega_k t} - e^{-i\omega_k t}) \sin(\theta_k/2) \cos(\theta_k/2) f_{-k}(0) (-e^{-i\omega_k t} + e^{i\omega_k t}) \sin(\theta_k/2) \cos(\theta_k/2) f_{-k}^\dagger(0) \right\rangle$$

$$= \left\langle \sum_k 2 \sin \omega_k t \sin(\theta_k/2) \cos(\theta_k/2) f_{-k}(0) 2 \sin \omega_k t \sin(\theta_k/2) \cos(\theta_k/2) f_{-k}^\dagger(0) \right\rangle$$

$$= \sum_k \sin^2 \theta_k \sin^2 \omega_k t$$

$$\begin{aligned} \sum_n c_n^\dagger c_n &= \sum_n \frac{1}{\sqrt{N}} \sum_k e^{ink} f_k^\dagger \frac{1}{\sqrt{N}} \sum_{k'} e^{-ink'} f_{k'} \\ &\Rightarrow \sum_{kk'} \frac{1}{N} f_k^\dagger f_{k'} \sum_n e^{in(k-k')} \\ &\Rightarrow \sum_{kk'} \frac{1}{N} f_k^\dagger f_{k'} N \delta_{kk'} \Rightarrow \sum_k f_k^\dagger f_k \end{aligned}$$

$$e^{i\omega_k t} - e^{-i\omega_k t} = 2 \sin \omega_k t$$

The Free Energy per site

$$\begin{aligned}
 f &= - \lim_{N \rightarrow \infty} \frac{\ln Z}{\beta N} = - \lim_{N \rightarrow \infty} \frac{\ln \text{tr} e^{-\beta \hat{H}}}{\beta N} = - \lim_{N \rightarrow \infty} \frac{\ln \text{tr} e^{-\beta \sum_k \omega_k (c_k^\dagger c_k - \frac{1}{2})}}{\beta N} = - \lim_{N \rightarrow \infty} \frac{\ln \text{tr} (e^{-\beta \omega_1 (c_1^\dagger c_1 - \frac{1}{2})} \cdot e^{-\beta \omega_2 (c_2^\dagger c_2 - \frac{1}{2})} \cdot \dots)}{\beta N} \\
 &= - \lim_{N \rightarrow \infty} \frac{\ln \text{tr} e^{-\beta \omega_1 (c_1^\dagger c_1 - \frac{1}{2})} \cdot \text{tr} e^{-\beta \omega_2 (c_2^\dagger c_2 - \frac{1}{2})} \cdot \dots}{\beta N} = - \lim_{N \rightarrow \infty} \frac{\ln \text{tr} e^{-\beta \omega_1 (c_1^\dagger c_1 - \frac{1}{2})} + \ln \text{tr} e^{-\beta \omega_2 (c_2^\dagger c_2 - \frac{1}{2})} \cdot \dots}{\beta N} \\
 &= - \lim_{N \rightarrow \infty} \frac{\sum_k \ln \text{tr} e^{-\beta \omega_k (c_k^\dagger c_k - \frac{1}{2})}}{\beta N} = - \lim_{N \rightarrow \infty} \frac{\sum_k \ln (e^{\frac{1}{2}\beta \omega_k} + e^{-\frac{1}{2}\beta \omega_k})}{\beta N} = - \lim_{N \rightarrow \infty} \frac{\sum_k \ln 2 \cosh \frac{\beta \omega_k}{2}}{\beta N} \\
 &= - \frac{\ln 2}{\beta} - \lim_{N \rightarrow \infty} \frac{\sum_k \ln \cosh \frac{\beta \omega_k}{2}}{\beta N} = - \frac{\ln 2}{\beta} - \frac{1}{2\pi\beta} \int_{-\pi}^{\pi} \ln \cosh \frac{\beta \omega_k}{2} dk
 \end{aligned}$$

The Internal Energy per site

$$\begin{aligned}
 u &= \frac{\partial(f\beta)}{\partial\beta} = \frac{\partial\left(-\frac{\ln 2}{\beta} - \frac{1}{2\pi\beta} \int_{-\pi}^{\pi} \ln \cosh \frac{\beta\omega_k}{2} dk \cdot \beta\right)}{\partial\beta} = \frac{\partial\left(-\ln 2 - \frac{1}{2\pi} \int_{-\pi}^{\pi} \ln \cosh \frac{\beta\omega_k}{2} dk\right)}{\partial\beta} \\
 &= \frac{-\partial\left(\frac{1}{2\pi} \int_{-\pi}^{\pi} \ln \cosh \frac{\beta\omega_k}{2} dk\right)}{\partial\beta} = \frac{-\left(\frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{\omega_k}{2} \frac{\sinh \frac{\beta\omega_k}{2}}{\cosh \frac{\beta\omega_k}{2}} dk\right) \partial\beta}{\partial\beta} = -\frac{1}{4\pi} \int_{-\pi}^{\pi} \omega_k \tanh \frac{\beta\omega_k}{2} dk
 \end{aligned}$$

$$\begin{aligned}
 u &= \frac{\partial(f\beta)}{\partial\beta} = \frac{\partial\left(-\frac{\ln 2}{\beta} - \lim_{N \rightarrow \infty} \frac{\sum_k \ln \cosh \frac{\beta\omega_k}{2}}{\beta N} \cdot \beta\right)}{\partial\beta} = \frac{1}{N} \frac{\partial\left(-\ln 2 - \lim_{N \rightarrow \infty} \sum_k \ln \cosh \frac{\beta\omega_k}{2}\right)}{\partial\beta} = \frac{1}{N} \frac{\partial\left(-\lim_{N \rightarrow \infty} \sum_k \ln \cosh \frac{\beta\omega_k}{2}\right)}{\partial\beta} \\
 &= \frac{1}{N} \frac{-\lim_{N \rightarrow \infty} \sum_k \frac{\omega_k}{2} \frac{\sinh \frac{\beta\omega_k}{2}}{\cosh \frac{\beta\omega_k}{2}} \partial\beta}{\partial\beta} = -\frac{\lim_{N \rightarrow \infty} \sum_k \omega_k \tanh \frac{\beta\omega_k}{2}}{2N}
 \end{aligned}$$

The specific heat

$$\begin{aligned}c_v &= \frac{\partial u}{\partial T} = \frac{\partial u}{\partial \beta^{-1}} = -\frac{\partial u}{\beta^{-2} \partial \beta} = -\beta^2 \frac{\partial \left(-\frac{1}{4\pi} \int_{-\pi}^{\pi} \omega_k \tanh \frac{\beta \omega_k}{2} dk \right)}{\partial \beta} = -\beta^2 \frac{\left(-\frac{1}{4\pi} \int_{-\pi}^{\pi} \omega_k \frac{\omega_k}{2} \left(1 - \tanh^2 \frac{\beta \omega_k}{2} \right) dk \right) \partial \beta}{\partial \beta} \\&= -\beta^2 \frac{1}{4\pi} \int_{-\pi}^{\pi} \frac{\omega_k^2}{2} \left(1 - \tanh^2 \frac{\beta \omega_k}{2} \right) dk\end{aligned}$$

$$\begin{aligned}c_v &= \frac{\partial u}{\partial T} = \frac{\partial u}{\partial \beta^{-1}} = -\frac{\partial u}{\beta^{-2} \partial \beta} = -\beta^2 \frac{\partial \left(-\frac{\lim_{N \rightarrow \infty} \sum_k \omega_k \tanh \frac{\beta \omega_k}{2}}{2N} \right)}{\partial \beta} = \beta^2 \frac{\left(\lim_{N \rightarrow \infty} \sum_k \frac{1}{2N} \frac{\omega_k^2}{2} \left(1 - \tanh^2 \frac{\beta \omega_k}{2} \right) \right) \partial \beta}{\partial \beta} \\&= \lim_{N \rightarrow \infty} \frac{\beta^2}{4N} \sum_k \omega_k^2 \left(1 - \tanh^2 \frac{\beta \omega_k}{2} \right)\end{aligned}$$

The Magnetization

$$\omega_k = 2 \sqrt{\left[\frac{h_z}{2} - \lambda \cos k\right]^2 + [\gamma \sin k]^2}$$

$$\begin{aligned} m &= \lim_{N \rightarrow \infty} \frac{1}{2N} \sum_k \frac{\partial \omega_k}{\partial h_z} \tanh \frac{\beta \omega_k}{2} = \lim_{N \rightarrow \infty} \frac{1}{2N} \sum_k \frac{\partial 2 \left(\left[\frac{h_z}{2} - \lambda \cos k\right]^2 + [\gamma \sin k]^2 \right)^{\frac{1}{2}}}{\partial h_z} \tanh \frac{\beta \omega_k}{2} \\ &= \lim_{N \rightarrow \infty} \frac{1}{2N} \sum_k \frac{\frac{1}{2} \cdot 2 \left(\left[\frac{h_z}{2} - \lambda \cos k\right]^2 + [\gamma \sin k]^2 \right)^{-\frac{1}{2}} \cdot 2 \left(\frac{h_z}{2} - \lambda \cos k \right) \cdot \frac{1}{2} \partial h_z}{\partial h_z} \tanh \frac{\beta \omega_k}{2} \\ &= \lim_{N \rightarrow \infty} \frac{1}{2N} \sum_k \left(\left[\frac{h_z}{2} - \lambda \cos k\right]^2 + [\gamma \sin k]^2 \right)^{-\frac{1}{2}} \left(\frac{h_z}{2} - \lambda \cos k \right) \tanh \frac{\beta \omega_k}{2} \end{aligned}$$